Garbage Collectors

Every Hadamard test throws away half its data. We recycle it.

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Qiskit Hackathon 2026 — Topic 5

after Faehrmann, Eisert & Kueng, *PRL* **135**, 150603 (2025)

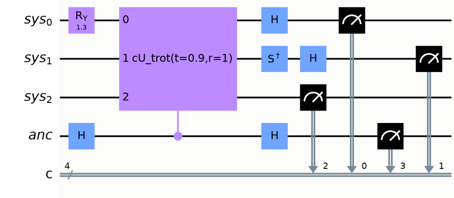
14 August 2026

Every Hadamard test throws away half its data

The standard Hadamard test measures **one ancilla qubit** and **discards the system register**.

Measure that "garbage" in *random bases* instead and it becomes a **classical shadow** — the same shots now also carry the system's observables.

We implement this end to end, validate every estimate against exact references, *quantify what it costs*, and characterise where it breaks.



3 system qubits + 1 ancilla

One circuit, one set of shots, seven physical quantities

The ancilla alone gives the interference signal:

$$\langle Z_a \rangle_\phi \;=\; \mathrm{Re}\Big[e^{i\phi}\,\mathrm{Tr}(U\rho)\Big]\,, \qquad \phi = 0 \to \mathrm{Re}\,\chi, \quad \phi = -\tfrac{\pi}{2} \to \mathrm{Im}\,\chi$$

Tracing out the ancilla leaves the *unweighted* shadow target, and weighting each snapshot by the ancilla outcome $a = \pm 1$ leaves the *joint* one:

$$\rho^{(I)} = \tfrac{1}{2}(\rho + U\rho U^\dagger), \qquad \mathbb{E}\big[a\,\hat{P}\big] = \mathrm{Re}\Big[e^{i\phi}\mathrm{Tr}(P\,U\rho)\Big]$$

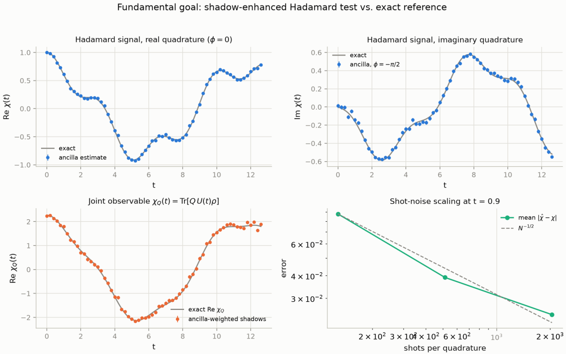
From **one** record set: $\chi(t)$, $\chi_Q(t)$, $\chi_H(t)$, $\chi_{Z_0}(t)$, $\langle H \rangle$, $\langle Q \rangle$, $\langle Z_0 \rangle$ [R009]

3,456 circuits · **256,000 shots** — one experiment, not seven.

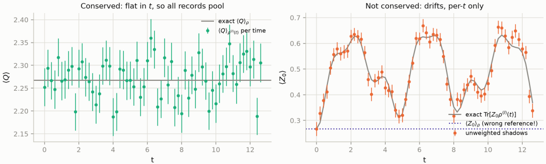
Every estimator agrees with theory — and the error bars are honest

- **56/56 checkpoints pass**, zero hard failures, zero advisory warnings.
- $\chi(t)$ rms error **0.0290** vs predicted sem **0.0277** — ratio **1.05**, so the error bars neither over- nor under-state. [R010]
- Shot-noise slope -0.463 vs theory -0.5: unbiased, statistics-limited. [R010]

The single-shot second moment is exactly $3^{w(P)}$, which predicts the measured error bars to **within 2%**. [R029]



Conserved quantities come out flat — from the very same shots



$\langle H\rangle = \mathbf{+0.0714 \pm 0.0082}$ (exact $+0.0739$)

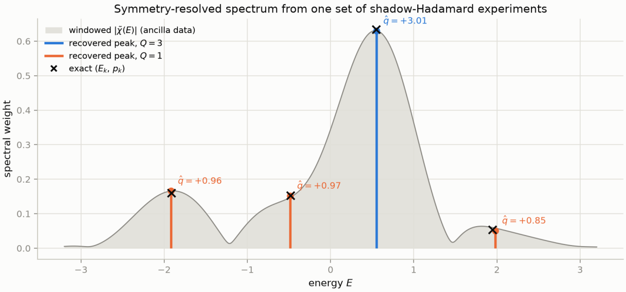
$\langle Q\rangle = \mathbf{+2.2708 \pm 0.0052}$ (exact $+2.2675$)

Pooled over all 128 settings, extracted from the *discarded* register of the $\chi(t)$ experiment. [R010]

Per-time $\langle Q\rangle$ is flat within 5σ at every one of the 64 time points (max 2.2σ). [R010]

No extra circuits (≤ 2 extra 1-qubit gates per system qubit); the statistical cost is on slide 7.

We recover the spectrum *and* label each line by symmetry sector



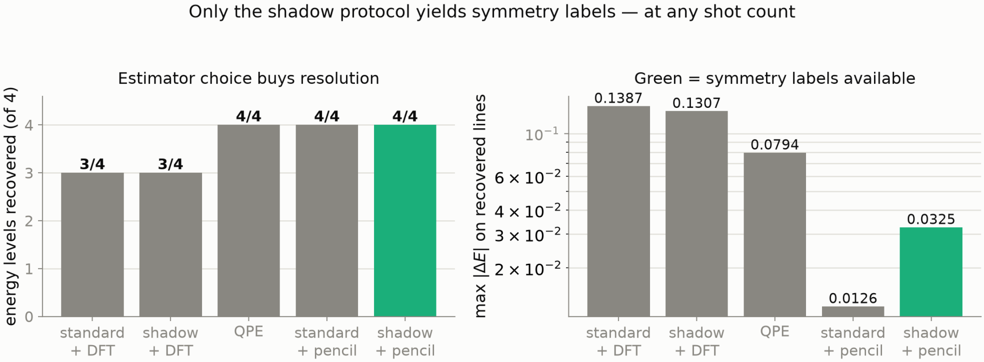
All 4 populated levels recovered: max $|\Delta E| =$ **0.033**, max $|\Delta p| =$ **0.006**, **every label correct**. [R012, R013]

Reproducible: **12/12 independent seeds** got all four labels right. [R019]

The colour is the deliverable. It is information the grey curve does not contain — and it cost *no additional circuits*.

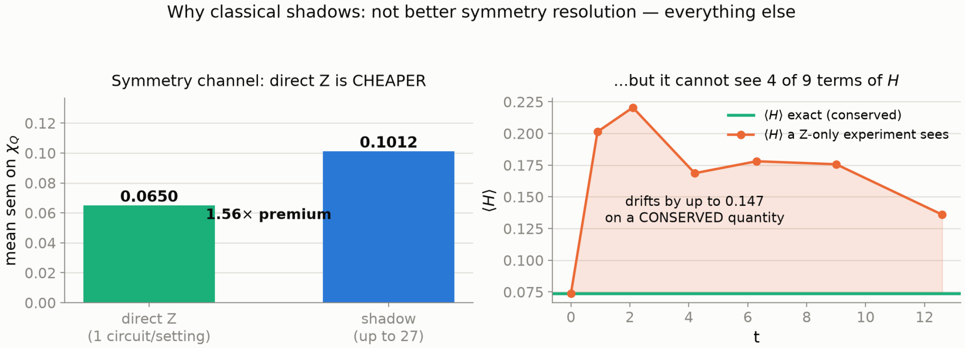
The grey band is the plain windowed Fourier transform. A better estimator sharpens it; *no* estimator can colour it.

What each choice actually buys



Estimator choice buys resolution ($3/4 \rightarrow 4/4$ lines). **Protocol** choice buys the labels — and nothing else can. [R011, R012, R021, R030]

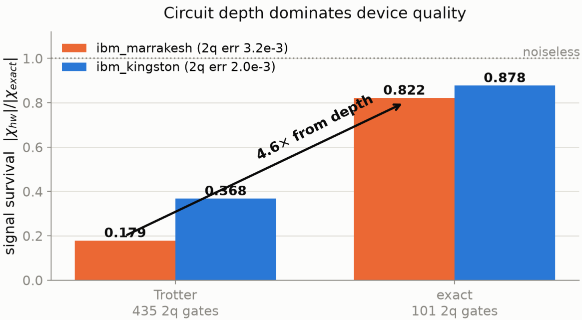
Do you even need shadows? We checked — and the answer is nuanced



$Q = \sum_j Z_j$ is *diagonal* in the computational basis, so a plain Z readout measures it directly — **1.56× cheaper** than our shadow estimator. [R039]

But a Z-only experiment sees **5 of 9** terms of H; the 4 it misses are the hopping terms, each with the *largest* coefficient. $\langle H \rangle$ is conserved — yet what it reports **drifts by 0.147**, and it has no second channel to notice.

It survives on real hardware once the circuit is shallow enough



Complete 2×2 grid. Depth dominates (**2.4–4.6**$\times$) over device choice (1.1–2.1$\times$). [R008, R018]

With a *full random-basis ensemble* on hardware: χ survival **0.962**, and a valid $\langle Q \rangle$ at **0.957** — the core claim, on a real QPU. [R023]

Caveat, stated plainly: one run per configuration. An independent run of an identical circuit 30 minutes later differed by $> 2\times$. Hardware varies; we report that. [R026, R031]

The honest cost: 13 experiments replaced for a 1.07$\times$ variance premium

The per-shot Pauli estimator and its exact second moment:

$$\hat{P} \;=\; 3^{\,w(P)} \prod_{j\in\mathrm{supp}(P)} s_j\, \mathbf{1}[b_j = P_j]\,, \qquad \mathbb{E}[\hat{P}^2] \;=\; 3^{\,w(P)}$$

Conventional route: **13 dedicated** modified Hadamard tests ($Q{=}3$, $H{=}9$, $Z_0{=}1$ Pauli terms).

Shadow route: **0 extra circuits** ($\leq$ 2 extra 1q gates per qubit).

Price paid instead — measured, not asserted:

$$\frac{\mathrm{sem}_{\text{shadow}}}{\mathrm{sem}_{\text{dedicated}}} = \frac{0.1012}{0.0949} = \mathbf{1.07}\times$$

on χ_Q specifically. [R021]

The 3^w model predicts the observed error bars to **within 2%**. [R029]

Reproduce it in under five minutes

- **37 sourced results rows** — every number maps to the command that produced it (`EVIDENCE.md`).
- Quickstart reproduces our headline *table* from saved data; the money plot ships pre-rendered.
- `hardware_run.py` reproduces our hardware arm in 3 commands.
- `BUGLOG.md` — including the numbers we got wrong and retracted.

QR CODE
TBD-NAME: paste repo QR here

First open implementation of PRL **135**, 150603. Prior-art search 2026-08-13; scope and limits recorded in `deck/PRIOR_ART_SEARCH.md`.

What we could not do, and where it breaks

- The windowed DFT **never** resolves the 1.031-spaced pair at *any* $T_{\max}$ we tested; the matrix pencil resolves it from $T_{\max} \geq 7.0$. A method limit, not a shot-count limit. [R011, R022]
- Exact-synthesis shallowness is an $n=3$ **artefact** — the advantage falls $3.41\times \to 1.16\times$ from $n=3$ to $n=4$ and is *expected* to invert beyond (measured at $n=3,4$ only). Not a scalable recipe. [R025]
- One noise study came out **confounded**; we exclude it rather than report it. [R017]
- Hardware supports *robustness claims only* — never a headline number. All Part A/B results are simulator.

We make no claim of outperforming classical computation.
The system is 3 qubits; a laptop diagonalises it instantly. It is small *on purpose*, so every estimate can be graded against an exact answer.

Team

- **TBD-NAME** — TBD-CONTRIBUTION
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Thank you — questions welcome.

Backup slides

Backup — the standard and shadow protocols give *identical* χ

Measured, not assumed: a genuine textbook Hadamard test over the full 64-point grid at identical budget gives

$$\mathrm{rms}_{\mathrm{std}} = 0.0265, \quad \mathrm{rms}_{\mathrm{shadow}} = 0.0290, \qquad \text{mean sem } 0.0277 \text{ vs } 0.0277$$

with $|\chi_{\mathrm{std}} - \chi_{\mathrm{shadow}}|$/combined sem: max 2.10σ, mean 0.88σ — statistically the **same random variable**. [R021]

So measuring the garbage register costs the ancilla signal *nothing*, and the two axes factorise cleanly: estimator $\to$ resolution, protocol $\to$ labels.

Backup — a self-validating decoherence witness

For a pure input state, exactly:

$$\mathrm{Tr}\left[(\rho^{(I)})^2\right] \;=\; \tfrac{1}{2}\left(1 + |\chi(t)|^2\right)$$

Left side from the **system register alone** (shadow snapshot pairs); right side from the **ancilla alone**. Two independent channels of the same experiment, so their gap certifies decoherence with **no exact diagonalisation and no classical simulation** — it survives to sizes where nothing can be verified.

ideal simulator	-0.1σ	(correct null)
noisy sim, $1\times$	-7.6σ	(fires)
noisy sim, $5\times$	-17.2σ	(dose-response)
real hardware (cleanest run)	-1.4σ	(correct null)

[R033]

Backup — future work, already prototyped: density of states

The DOS is the Fourier transform of the *trace* of the propagator:

$$G(t) \;=\; \frac{1}{\sqrt{2\pi}}\,\mathrm{Tr}\Big[e^{-i\hat{H}t}\Big]$$

(Goh & Koczor, arXiv:2407.03414v2, Eq. 7)

Our $\chi(t) = \mathrm{Tr}[U(t)\rho]$. Set $\rho = \mathbb{1}/2^n$ and **the same experiment becomes a DOS estimator** — one line changed, analysis chain untouched.

Prototyped: peak weights correctly become *degeneracies* (isolated levels 0.11–0.13 vs predicted 1/8; a 3-fold merged peak 0.354 vs predicted 3/8). [R038]

Where we could improve on that paper: it obtains fixed-particle-number DOS with *one experiment per sector*. Our shadow channel measures $\langle Q^k\rangle$ from the *same* shots, so the charge moments at each peak recover **all sectors from a single run**. Feasibility checked: Q^2, Q^3 are 4 Pauli terms each; moment inversion condition number 31.

Backup — two bugs we caught, and how

B04 — a fixed-basis shadow estimator silently returns a *different* observable. The estimator is unbiased only for i.i.d.-uniform bases; with one fixed basis the indicator is deterministic and non-matching terms contribute structurally zero. We reported a hardware $\langle Q \rangle$ this way and blamed noise. On a *noiseless* simulator the same code is **65σ off** — so it was never noise. Retracted; `hardware_run.py` now refuses to print system observables without `--shadow`.

B06 — an error bar computed over the wrong units. Purity is a U-statistic over *pairs* of snapshots; we used $\mathrm{std}/\sqrt{n_{\mathsf{pairs}}}$, but pairs share shots, so the effective sample size is the number of *shots*. That faked a 2.2σ signal. **Only the ideal-simulator control leg exposed it** — the hardware number alone looked fine.

A witness validated only where the truth is unknown cannot be trusted.